



East West University
Department of Computer Science and Engineering
BSc in Computer Science and Engineering Program
Lab Project

Course: CSE 110, Object Oriented Programming
Instructor: Sadia Nur Amin, Senior Lecturer, CSE

Objective: Develop a Java-based application that demonstrates your understanding of core object-oriented programming principles, including classes, inheritance, encapsulation, polymorphism, and collections. You will choose a project topic that interests you and implement it based on the following requirements.

Requirements:

1. Class Structure: (minimum 4)

- Define Classes: Create at least three core classes for your application. Ensure that each class has meaningful attributes and methods that represent its responsibilities.
- Encapsulation: Use private attributes and provide public getter and setter methods as needed to encapsulate data.

2. Inheritance and Polymorphism:

- Inheritance: Create at least one base class (abstract or concrete) and at least one subclass that extends it. Ensure that the subclass inherits and possibly overrides methods from the base class.
- Polymorphism: Demonstrate method overriding and/or method overloading in your classes. Implement a polymorphic behavior where appropriate.

-Abstract/ Interface

3. Collections:

- Manage Data: Use Java collections (e.g., `ArrayList`, `HashMap`, etc.) to store and manage data within your application. Ensure that your application can efficiently handle the data and perform required operations.

4. Exception Handling (minimum 3 types):

- Error Management: Implement proper exception handling to manage errors that may occur during runtime. This could include handling invalid inputs, missing data, or other potential issues.

5. User Interface: Bonus Challenge (Optional):

- Interaction: Provide a simple text-based or graphical user interface to interact with your application. This could involve using the console, Swing, JavaFX, or any other UI library of your choice.

6. UML Diagram of the project

Topic Choice: Choose a project topic that interests you and aligns with the requirements.

Submission Guideline:

- Provide well-documented source code, including comments and descriptions of your classes and methods.
- Include the **UML diagram** of the project.

Project Demonstration and viva:

- Will be taken during lab hours.